

Received 24 August 2024, accepted 27 September 2024, date of publication 3 October 2024, date of current version 22 October 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3472012

RESEARCH ARTICLE

PDCNET: Deep Convolutional Neural Network for Classification of Periodontal Disease Using Dental Radiographs

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This work was supported in part by the National Natural Science Foundation of China under Grant 62262019; and in part by Hainan Provincial Natural Science Foundation of China under Grant 621RC1059, Grant 621MS038, and Grant 823RC488.

ABSTRACT Dental health plays a pivotal role in determining the general health and quality of life of a person. Dental caries is a prevalent health problem in the world; therefore, prompt and effective dental caries treatment is necessary for individuals who require pain relief. Conventional methods of diagnosing dental diseases, like visual examination and radiographic testing, depend on qualified medical professionals and can be incredibly labor-intensive and imprecise in their diagnosis. To overcome these challenges, this study designed a deep learning (DL) model for the diagnosis of several dental conditions such as tooth decay, non-periodontal, and periodontal disease. A novel model named periodontal disease classification network (PDCNET) based on convolutional neural network (CNN) has been developed for the identification of periodontal disease using dental radiographs. Additionally, the proposed PDCNET model has been evaluated on two publicly available benchmark datasets of dental caries. To handle the imbalanced classes of the dental caries dataset, this study used the SMOTE TOMMEK method to generate new synthetic samples for the minority categories to ensure the periodontal disease dataset is balanced. The proposed PDCNET model acquired a 99.79% AUC, 98.39% recall, 98.39% accuracy, 98.39% precision, and an F1-score of 98.31%. Furthermore, the performance of the proposed PDCNET model is contrasted using the six baseline pre-trained classifiers such as EfficientNet-B0 (M₁), DenseNet-201 (M₂), Vgg-16 (M₃), Vgg-19 (M₄), Inception-V3 (M₅), and MobileNet (M₆) in terms of many parameters. The levels of accuracy attained by M₁, M₂, M₃, M₄, M₅, and M₆ are 85.94%, 94.37%, 91.96%, 93.57%, 89.95%, and 94.77%, respectively. The findings showed that the PDCNET model has superior outcomes as compared to baseline models and provides significant assistance to the dentist in the diagnosis of dental disease.

INDEX TERMS Deep learning, periodontal disease, tooth decay, dental radiographs.

The associate editor coordinating the review of this manuscript and approving it for publication was Roberta Palmeri.

I. INTRODUCTION

Oral illness harms almost 50% of the population of every country in the world, and approximately 2.3 billion people are suffering from the effects of chronic dental caries [1], [2], [3]. This illness known as dental caries, more commonly referred

to as tooth decay, is an illness that is marked by the flaw to the teeth that is caused by bacteria in the mouth that develop lactic acids. These acids directly attack the enamel layer of the tooth, which is why dental caries is also known as tooth decay. This can, over time, result in a tiny hole around the teeth, that, if not treated, could lead to toothache, being infected, and the loss of teeth [4], [5], [6], [7]. Some lesions of caries cannot be recognized by sight or touch. To enhance the detection rate, imaging techniques are utilized [5], [6], [7], [8], [9], [10]. Despite the use of medical imaging, in the teeth of the patient the dental surgeon might have missed the initial caries lesions. Because of this, the effectiveness of the diagnosis and treatment is dependent not only on the capabilities of the technology but also on the individual abilities of the reading specialist [11]. In addition to this, the examination of X-ray images adds more work for the attending physician. An automatic detection system that has a high level of consistency and precision is required to provide clinical stomatologists with assistance in the objective diagnosis of caries.

Techniques such as visible light transillumination [12], another calibrated diaphragm computed tomography [13], [14], and the globally recognized cavities diagnosing and evaluation setup [15] are examples of these, and some of the techniques like quantitative light-induced fluorescence that were utilized in the early research on dental caries detection. A separate study [16] developed a strategy for improving panoramic dental radiography to improve its accuracy as a diagnostic tool for cavities of teeth through the use of methods of image processing. In addition, the scientists examine computer-assisted cavity recognition utilizing X-ray photographs [17], [18], [19], [20], [21]. The study [19] examined and confirmed the effectiveness of a density analysis auxiliary tool in assisting dental specialists in the detection and classification of caries based on user feedback. Researchers achieved this with the Logicon Caries Detector (LCD) which utilizes improved CAD software. The study [20] developed a neural network called artificial feed-forward to identify dental lesions in panoramic dental X-ray pictures. The results of their analysis showed that the accuracy of their classification was 98.7%. An unsupervised learning system for the diagnosis of caries based on clump identification and cluster analysis was developed by [22]. This method worked quite well, with an accuracy rate of 96% in the diagnosis. In the study [23], they designed a step-by-step algorithm for diagnosing interproximal caries as well as enamel by using radiography in digital periapical pictures as their data source.

In recent years, deep learning (DL) has emerged as a major topic in the field of dentistry, specifically in the identification of dental cavities [20], [21], [22], [23], [24], [25]. Because there haven't been a lot of studies done on the issue of "deep learning-based caries classification," it is still something that has to be investigated further. The author [14] designed a CNN based on the classification of tooth decay and they achieved the classification accuracy of 94.60%. The limitation of their method is that it cannot preserve the borderline among two teeth if the technique of simplification detailed for

that method corrupts the boundary. Wisdom teeth of different manifestations along every tooth meshing are responsible for a sizeable amount of the mistake. This is because the wisdom teeth are either not present, not fully formed, or only partially visible in the mouth. The training process is challenging due to the limited amount of data available on wisdom teeth. Caries are found in a significantly higher percentage of children and teenagers as they get older. Following puberty, there is an increase in the prevalence of periodontal disease, which demonstrates an epidemiologic trait of being softly maintained after teenagehood [11], [12], [13], [14].

Since dental caries in young children can spread to the adult teeth and cause other oral health problems if left untreated, they must be addressed as soon as possible. As a result, we are driven to develop a model that can identify periodontal disease, non-periodontal disease, and tooth decay using dental radiographs, to prevent the adults from severe infection. This study develops a novel model, named a deep learning-based periodontal disease classification network (PDCNET) which is based on a CNN for the classification of periodontal disease, non-periodontal disease, and tooth decay using dental radiographs. This is the primary study that uses DL methods for the classification of dental diseases. To handle the imbalance class problem of the datasets, we applied the synthetic minority oversampling technique (SMOTE) Tomek with the PDCNET model. Additionally, the PDCNET model is also compared with six baseline classifiers such as EfficientNet-B0 (M_1) [5], DenseNet-201 (M_2) [6], Vgg-16 (M_3) [7], Vgg-19 (M_4) [8], Inception-V3 (M_5) [9], and MobileNet (M_6) [10].

The following are the key contributions of this work:

1. The novel CNN-based proposed PDCNET model is designed to identify periodontal disease, non-periodontal disease, and tooth decay using dental radiographs. We can extract dominating features from dental radiographs using the PDCNET model, which can help with dental disease identification. Additionally, we simplify the proposed PDCNET model by reducing the total number of trainable parameters to develop a robust classifier.

2. The class disparity issue is causing havoc on medical datasets and is contributing to the CNN model's decreasing accuracy. To increase the accuracy of the PDCNET model, we gather sample mixtures from the dental radiographs at each class using an upsampling technique called SMOTE Tomek. Furthermore, we applied the Grad-CAM heat map to clearly show all the characteristics associated with periodontal disease.

3. The PDCNET outperformed six baseline classifiers, including M_1 , M_2 , M_3 , M_4 , M_5 , and M_6 , based on a variety of evaluation metrics, including accuracy (ACU), an area under the curve (AUC), precision (PRE), recall (REC), loss, and F1-score. Moreover, the proposed PDCNET model yielded noticeable and significant results when compared with the state-of-the-art (SOTA) methods.

The study is organized as follows: Section II presents the modern literature on dental disease classification including

periodontal disease, non-periodontal disease, tooth decay, etc. Section III explains the dataset description, data pre-processing, SMOTE Tomek, proposed PDCNET model, and performance evaluation metrics. Section IV presents the findings and discussions of the experiments. The study has been concluded in Section V.

II. LITERATURE REVIEW

In dentistry, computer-aided image classification has been introduced to make use of diversification for different procedures turned out to be described in the literature [26], [27], [28], [29]. DL has been successfully used for computer vision challenges, including image categorization, unmanned surface vehicles (USVs) [26], electric marine surface vehicle fleets [27], and object detection. In recent years, there has been a rise in the popularity of using algorithms that are based on DL to do image classification. Table 1 presents the modern literature on the use of DL approaches for dental disease classification. Jader et al. [30] designed a CNN-based model with mask region for the classification of tooth decay. They trained and tested their proposed model on a total of 1500 panoramic X-ray radiographs. Their proposed model achieved an accuracy of 92.14%, an F1-score of 96.02%, a recall of 88.00%, and a specificity of 92.12%.

In the study [31], they used a cross-validation procedure with four iterations employing a total of one hundred panoramic dental radiographs. As a direct consequence of this, the sensitivity of tooth detection was found to be 96.4%, while its accuracy was found to be 93.2%. An AlexNet-based model was utilized by [32] to classify dental cone-beam computed tomography (CBCT) representations of various tissues of the teeth. They trained the AlexNet model on a total of 42 dental radiographs and then tested it using 10 dental photographs, achieving an accuracy rate of 80%. In addition, the study [33] used the pre-trained model to recognize and count teeth in dental periapical videos by making use of speedier R-CNN characteristics. In this specific instance, 800 dental photos were utilized for the training process, 200 for the testing procedure, and 250 for the validation procedure. They utilized a neural network to be able to provide a correct evaluation and achieved a precision of 90.15%.

The researchers [34] used periapical images that have been treated with Faster RCNN and R-FCN (region-based fully convolutional networks) in their work. When training the network, a total of 700 dental photos were utilized; 200 photos were utilized for testing, and 100 photos were utilized to validate the results. This model's output seems to have a recall percentage of 96.10 as well as a precision of 95.78%. The study [35] used Faster R-CNN architecture in conjunction with 1352 panoramic photos to identify teeth. In their study, the researchers were able to attain a sensitivity of 0.9941 and a precision of 0.9945.

The RESNET18 and RESNEXT50 models have been implemented by [36] via the 10-fold cross-validation approach. During training, a one-cycle learning rate strategy

was used with a minimum learning rate of 0.0001 as well as a maximum learning rate of 0.01. To evaluate model performance, several distinct metrics, such as PPV/NPV and AUC, were applied. However, they achieved an AUC value of 0.92. By employment of CNN architecture and the PyBrain package, the authors [25] were successfully able to identify teeth. Their system had an overall accuracy of 0.93. The study [14] used an AlexNet model on the dental database consisting of 100 panoramic radiographs to classify teeth as molars, incisors, or premolars with an accuracy of 0.92 or higher.

The study [12] designed a deep-learning model for the classification of vertical root fracture (VRF) using panoramic radiography. Therefore, they collected 330 panoramic photos from the hospital imaging database that showed VRF teeth with evident fracture lines. From the total of 300 photographs, 240 were selected to serve as a training dataset, and 60 were chosen to serve as a testing dataset. Their model achieves a precision of 0.93, whereas the F-measure was 0.83.

The study [9] designed a highly optimal network named PraNet was established for the classification of tooth decay. Moreover, direct segmentation in real time was achieved by [17] using DeepLab. An improved model category of the UNet network was reported by [21] an author [29] for detecting dental infections based on X-ray images.

Mouth fitness has become an extremely important variable in observing comprehensive amounts of our lives as we have transitioned into an aging society. It is estimated that periodontal disease may have the 6th extreme level of repeated illness globally, making that become the greatest widespread condition affecting humans [15]. Park et al. [37] proposed a CNN-based model for detecting teeth disorder by using teeth frontal color optical images. They compared the performance of the proposed model with ResNet152. According to their findings, the suggested model performed 11.45% superior as compared to the ResNet152.

According to [38] the goal of the study was intent on whether or not AI models are effective in determining whether or not a region has had alveolar bone loss. They used the YOLOv5 model for performing the segmentation to find and label periodontal bone loss sites on panoramic images that are 685 in number. In these regions of the maxillary incisors, the F1-score and sensitivity values were highest, whereas the overall alveolar bone loss was related to the lowest values. This suggests that artificial intelligence has a significant amount of promise for assessing the severity of periodontal bone loss problems.

The goal of the study [39] was to construct three patient cohorts (disease improvement, disease progression, and no disease change) utilizing longitudinal electronic dental records (EDR) to track the evolution of periodontal disease (PD) over time. Additionally, two DL models have been to extract information from clinical notes. A retrospective analysis was performed on 28,908 patients who had received a comprehensive oral evaluation at Indiana University. A total

TABLE 1. Recent studies for the classification of periodontal disease using deep learning models.

Ref	Years	Model	Disease	Type	Results
[1]	2024	CNN	Periodontal Disease	Binary	Accuracy = 95.30%
[37]	2023	CNN	Oral and Maxillofacial Implantology.	Multi	Accuracy = 86.87%
[2]	2024	CNN	Periodontal Disease	Binary	Accuracy = 92.77%
[38]	2023	ResNet152 + CNN	Periodontal Disease	Binary	Accuracy = 94.49%
[3]	2024	ResNet + SAM	Periodontal Disease	Binary	F1-score = 88.60%
[39]	2023	YOLO-v5	Periodontal Bone Loss	Multi	F-Score = 75.00%
[40]	2023	CNN	Periodontal Disease	Multi	Accuracy = 92.00%
[41]	2023	CNN	Periodontal Disease	binary	Accuracy = 94.89%
[42]	2022	NASNet	Cavity, Tooth Decay, and Teeth filing	Multi	Accuracy = 96.00%
[30]	2018	CNN	Caries and Endodontic Disorders	Binary	Accuracy = 98.00%
[31]	2019	CNN	Restored teeth	Binary	Accuracy = 93.2%
[32]	2016	AlexNet	Tooth Segmentation	Binary	Accuracy = 91.0%
[33]	2019	R-CNN	Missing Teeth, Filled teeth, and Orthodontics	Multi	Accuracy = 92.12%.
[35]	2018	CNN	Missing Teeth, Cavities, and Tooth Decay	Multi	Precision = 95.8%
[43]	2019	Vgg-16	Missing Teeth, Cavities, and Tooth Decay	Multi	Sensitivity = 96.00%
[44]	2016	ANN	Missing Teeth, Cavities, and Tooth Decay	Multi	Accuracy = 93.50%
[45]	2017	AlexNet	Molar Mouth Gap, Premolar	Multi	Accuracy = 90.00%
[46]	2020	CNN	VRF	Multi	F-Score = 83.00%.
[47]	2017	FCN-SegNet	Tooth Decay	Binary	Accuracy = 90.19%
[48]	2020	CNN + YOLOv3	Jaws and Tooth Decay	Binary	Accuracy = 91.3%
[49]	2018	GoogleNet	Distal roots and Extra roots	Binary	Accuracy = 86.9%
[11]	2021	CNN	Maxillary Sinuses	Binary	Accuracy = 87.5%
[13]	2022	CNN	Periodontal disease	Binary	Accuracy = 81.00%
[15]	2021	DeNTNet	Periodontal Disease	Binary	F-score = 75.00%
[27]	2021	CNN	Root caries	Binary	Accuracy = 97.10%
[28]	2020	RF	Filling and Tooth decay	Binary	Accuracy = 94.29%
[17]	2022	ANN	Dental Caries	Binary	Accuracy = 92.00%
[18]	2022	YOLO	Cavity and Caries lesion	Multi	Accuracy = 87.40%

of 28,908 patients who had received a comprehensive oral evaluation at Indiana University School of Dentistry (IUSD) facilities between January 1, 2009, and December 31, 2014, were included in the retrospective analysis. In the study [40], a deep learning model was developed to assist general dentistry practitioners in effectively diagnosing periodontitis at an earlier stage. The accuracy of their model was determined to be 80.00%.

III. MATERIALS AND METHODS

This section comprises the approach of experimentation that is adopted for analyzing the performance of the proposed model along with six deep CNN classification methods. These deep CNN classification methods include EfficientNet-B0 (M_1), DenseNet-201 (M_2), Vgg-16 (M_3), Vgg-19 (M_4), Inception-V3 (M_5), and MobileNet (M_6).

A. FLOW OF PROPOSED MODEL FOR AN EARLY DENTAL DISEASE DIAGNOSIS

Cavities are the most widespread tooth issue caused by inadequate dental care. The identification of initial cavities enables preventative treatment, and X-ray images are an efficient method of diagnosis of initiated cavities in the teeth. A significant revolution has been caused in the field of healthcare and the medical sciences due to image processing. At present, image processing has an impact on practically every sector of the medical industry. It is not necessary to perform surgery for medical experts to conduct an examination of the internal organs of an individual's body at the time of the diagnostic phase.

A person is incapable of analyzing medical imaging exactly as a machine taking reliable findings from them. An automated system developed on a medical dataset is entirely suited to delivering accurate findings within a few seconds, while a team of doctors may require days to achieve the same conclusion. Computer vision and image processing techniques are essential components of today's medical care systems,

which is why these technologies are so crucial to these systems. Dental disease has recently become one of the diseases that is rising at the fastest rate worldwide. The community of researchers contributes significantly to the development of capable automated systems for efficient and reliable evaluation, as well as assists in making daily improvements [36], [50], [51].

In the process of this research, a fully automated system that is capable of identifying dental diseases has been designed. The input dataset is already processed with the assistance of normalization and a crucial technique of altering categorical variables of data to be established in the proposed model with the aid of the one-hot encoder. After that, an approach known as the Synthetic Minority Oversampling Technique (SMOTE) Tomek is applied to manage the issue of an unequal data set [42]. This approach oversamples the categories to equalize the dataset into balance. Then, the dataset is segmented into a training set, a validation set, and a test set by allocating 80%, 10%, and 10% of the data to each of those categories, respectively.

In addition, the features are taken out by making use of a conventional CNN to properly train the suggested model,

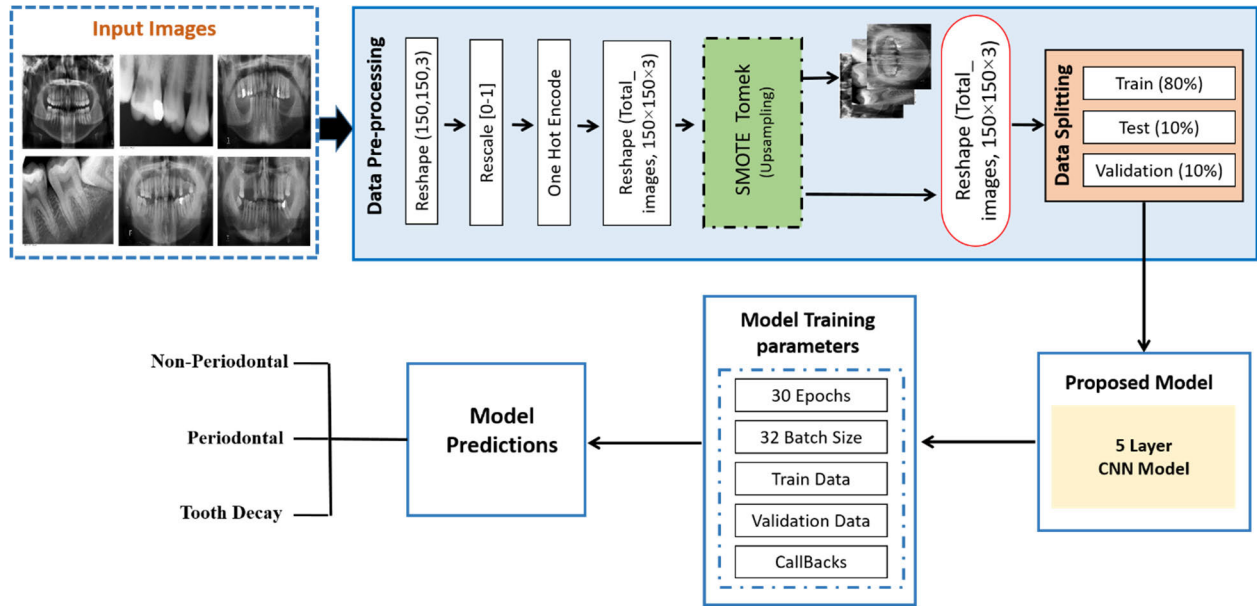


FIGURE 1. Study flow of the proposed model for the diagnosis of periodontal disease.

as can be seen in Figure 1. The amount of training parameters required for dental disease classification is significantly lower in contrast with [42], [52], and [53] for the level of reliability of the model. The category activating map is represented by the Grad-CAM heat-map process, which shows the qualities that are essential to the categorization of an image sample.

B. DATASET DESCRIPTION

There are several datasets related to dental diseases are available but this study mainly focuses on three different types of dental diseases such as class 0 (non-periodontal), class 1 (periodontal), and class 2 (tooth decay). Two datasets on dental diseases were gathered from different sources to train and evaluate the proposed model. Initially, we gathered 50 periodontal and 50 non-periodontal dental radiographs from a Kaggle repository [54]. Secondly, we collected 926 tooth decay dental radiographs from the source discussed in the study [55]. Table 2 shows the division of dental dataset categories before up-sampling. The sample size has been adjusted to 150×150 pixels. The iterations of the test experiment did not exceed 30 epochs. After the successful execution of all of the validation and training epochs, the proposed model attained the level of accuracy that was expected.

TABLE 2. Division of Dental Disease dataset classes before up-sampling.

Dental Diseases Classes	No. of Dental Radiographs
Non-Periodontal	50
Periodontal	50
Tooth Decay	926

C. TO EQUALIZE THE DENTAL DISEASE DATASET VIA SMOTE TOMEK

The two techniques that are usually employed in the procedure of re-sampling are oversampling and under-sampling.

We have utilized the SMOTE TOMEK technique for this research study that we are conducting. SMOTE produces new samples according to class closest neighbors, whereas TOMEK is a method that can be used to implement condensed closest neighbors. By raising the overall quantity of data included within the minority class, the employment of the SMOTE TOMEK method aids in ensuring that the number of samples in the classes remains balanced.

Both techniques operate in order, and SMOTE selects a random instance from a minority category and uses new samples to enhance its proportion. After that, Tomek chooses a sample at random and throws it out if any of its immediate neighbors are associated with a minority category.

Table 3 shows the division of dental dataset categories after the up-sampling. SMOTE TOMEK combines the instances of each kind in this manner and successfully resolves the database imbalance issue. Figure 2 displays images of the three main categories of dental disease after SMOTE TOMEK. The Nearest Neighbor technique is utilized by SMOTE TOMEK to generate new synthetic samples for the minority categories to ensure the database can be made more equal.

TABLE 3. Division of Dental Disease dataset classes after up-sampling.

Dental Diseases Classes	No. of Dental Radiographs
Non-Periodontal	925
Periodontal	912
Tooth Decay	926

D. PROPOSED MODEL COMPONENTS

In the following subsections, the primary components of the suggested model are presented briefly.

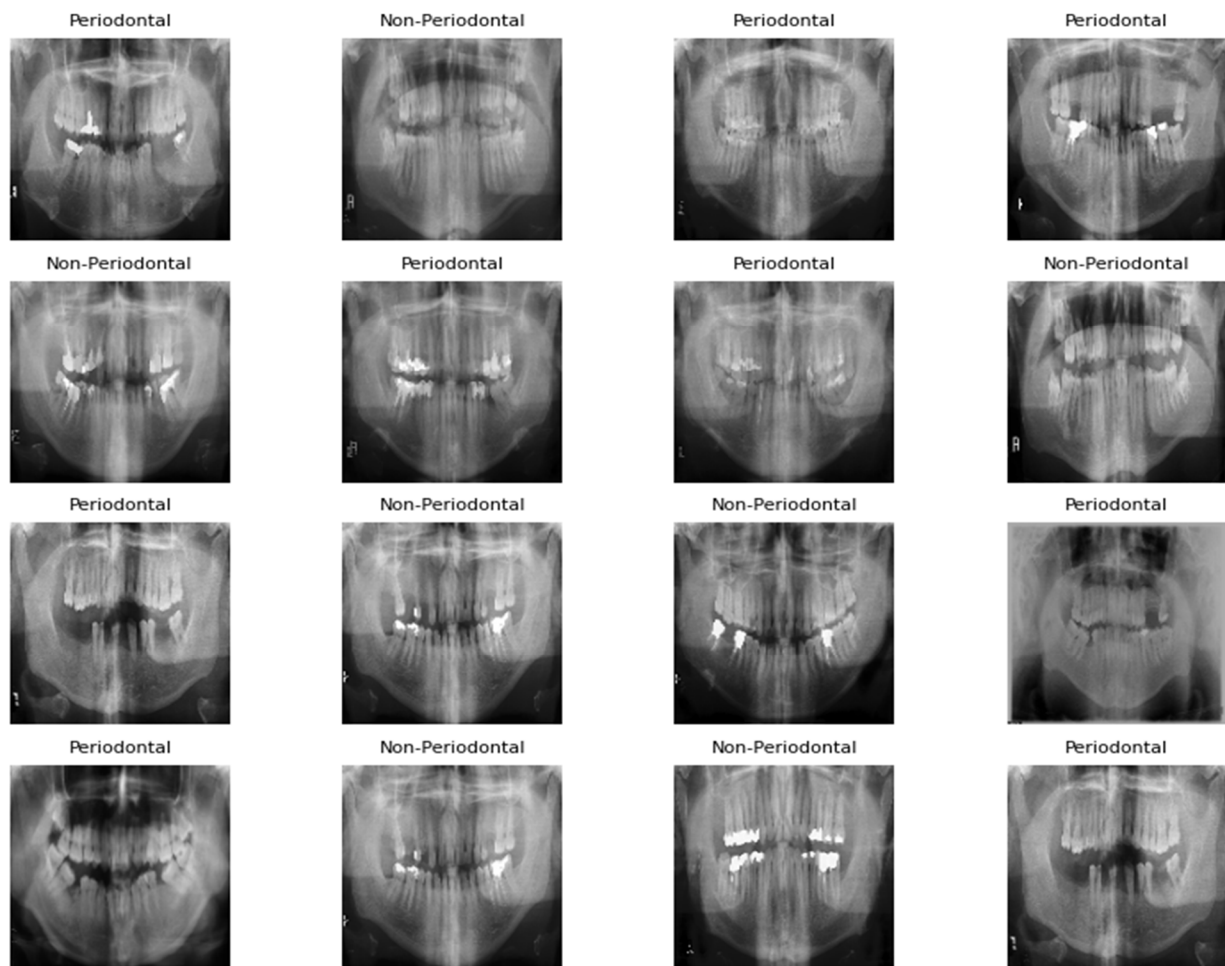


FIGURE 2. Dental disease data samples of images without up-sampling.

1) ARCHITECTURE OF THE PROPOSED MODEL

The structure of a CNN is similar to the connectivity layout of the brain of a human being. CNNs, similar to the human brain, are made up of billions of neurons that are arranged in a specific manner.

CNNs are a type of Deep Neural Network (DNN) that is capable of identifying and categorizing specific image features, and they are extensively employed for evaluating visual images. The identification of images and videos, the classification of images, image analysis for medical purposes, computer vision, and the processing of natural languages are among their many applications. In this study, we present a novel CNN model that is capable of precisely disease classification of dental disorders. The proposed model is constructed of 5 convolutional blocks, and every single convolutional block is set up with a MaxPool2D layer and activation function that is ReLU. After this, we used one layer of dropout, one dense block that consists of the ReLU function and kernel initializer, and then one dense layer and a SoftMax classification layer were applied for the output, as shown in Figure 3.

2) CONVOLUTIONAL BLOCKS OF PROPOSED MODEL

Every single convolutional block provides a conv2D, a ReLU, and a MaxPool2D, and the convolutional block is the most important establishing component of the model that was suggested. The CNN algorithms' use of the phrase "Convolution" relates to the computational equation for convolution, which is a particular form of linear calculation in which 2 functions are multiplied to generate a 3rd function that shows how the shape of one of the functions is influenced by another. The convolutional layer provides a set of filters, which are more often known as kernels. Both the height and width of the filter are reduced compared to the real dimensions of the image. A convolutional method that identifies and detects the multiple image features that are to be evaluated as a part of the process of feature extraction. The primary objective of the max-pooling layer is to reduce the total dimension of the condensed feature map to decrease the amount of computation that is required. The greatest element in max-pooling is derived from the feature map. Numerous activation functions are available, but we relied on ReLU as a consequence of its quick and time-efficient

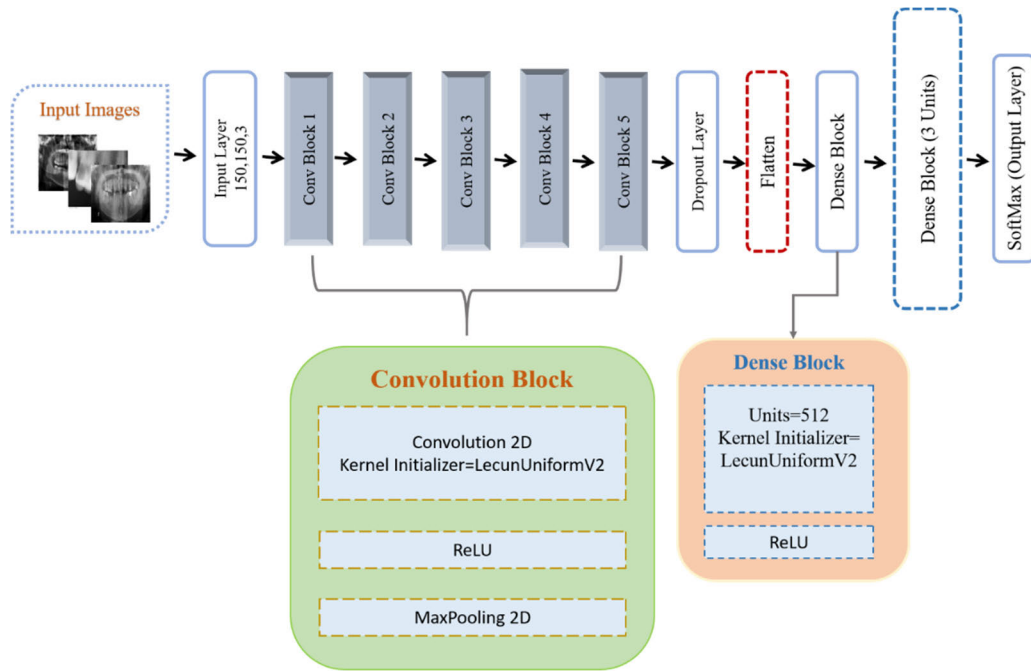


FIGURE 3. Entire structure of proposed model to diagnose dental disease.

evaluation. The network layout and model description of the suggested approach that was adopted for the categorization of dental conditions are presented in Table 4. The problem of vanishing gradients can be overcome by employing an activation function known as a rectified linear unit (ReLU). When ReLU is activated, all negative values are converted to zero.

The kernel initializer is responsible for the process of determining the weights for the 2D convolutional layer. The convolutional layers assemble characteristics alternately over time. Local designs like corners, segments, and slopes are detected in the earliest layers, and regional features are created with the assistance of these patterns.

3) DROPOUT LAYER

When every single feature is linked to the FCL, overfitting can occur in the training data. Overfitting happens if a model performs well enough on the training dataset but it negatively affects its performance when applied to new data [56]. To address this problem, a dropout layer has been implemented. This layer involves the drop of some of the neural connections in the neural network as it is being trained, which results in a more condensed model. When a dropout rate of 0.3 occurs, thirty percent of the neurons that comprise the neural network are randomly removed. The efficacy of a proposed model is enhanced by dropout, which simplifies the network to avoid overfitting. Neurons are dropped from the neural networks as they are trained [57]. These layers reduced the amount of time necessary for the model to be trained and the complexity of the network by shifting nodes on and off.

4) FLATTEN LAYER

The Flatten function transforms the feature map that has been pooled into one column that is then sent to the fully connected layer (FCL) [58]. This layer aims to compress multiple-dimensional input data to just one dimension. Figure 4 demonstrates how the flattened layer manipulates the feature map to ensure that it can be transmitted to the dense Layers. It is revealed that this layer serves as a connection for both the convolutional and dense layers. Convolution layers admit multidimensional types of data as input, whilst dense layers utilize only a single-dimensional type of data.

TABLE 4. A Brief Description of the Parameters that utilized by the PDCNET.

Layer Name	Output Shape	Param #
PDCNET_ Input_Layer_00	(None, 150, 150, 3)	0
PDCNET_ Conv-Block01	(None, 150, 150, 8)	224
PDCNET_ Conv-Block02	(None, 75, 75, 16)	1168
PDCNET_ Conv-Block03	(None, 37, 37, 32)	4640
PDCNET_ Conv-Block04	(None, 18, 18, 64)	18496
PDCNET_ Conv-Block05	(None, 9, 9, 128)	73856
PDCNET_ Dropout Layer	(None, 4, 4, 512)	0
PDCNET_ Flatten Layer	(None, 2048)	0
PDCNET_ Dense-Layer01	(None, 512)	1049088
PDCNET_ ReLu	(None, 512)	0
PDCNET_ Dense-Layer02	(None, 3)	2052
Output Layer (SoftMax)	(None, 3)	0
Total Params		1,149,524
Trainable Params		1,149,524
Non-Trainable Params		0

5) DENSE LAYER

This layer is recognized as the fully connected layer (FCL). The FCL contains biases and weights along with neurons and

is utilized to link neurons among two distinct layers [59]. The last few layers of the structure of CNN are comprised of these layers, which are typically positioned before the output layer. In this phase, the previous layer's input image is flattened and transmitted to the FCL. The vector of data is then given some more FCLs, where computation operations typically occur [60]. At this point in the process, the images contained within these layers are recognized and given a category labeling.

6) SOFTMAX ACTIVATION

Softmax is frequently employed as the activation for the final layer of a network for classification because the outcome can be analyzed as a distribution of probabilities [59]. Softmax activation is a supervised learning technique that is typically employed when there are various categories required. In other words, softmax is the activation function for classification issues with multiple classes requiring class membership on more than two category identifiers. The method known as SoftMax is implemented, having the total quantity of neurons being equivalent to the total amount of classes [56].

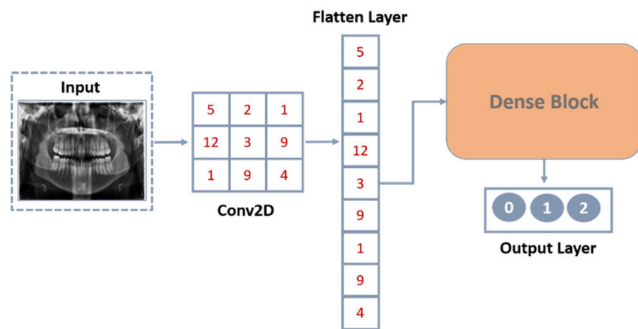


FIGURE 4. The steps for transferring the feature map via flattening it.

E. DL CLASSIFIERS

The following section comprises the six baseline DL classification methods that were utilized to accomplish the healthcare objective of dental condition categorization. These classification methods are commonly referred to as M_1 , M_2 , M_3 , M_4 , M_5 , and M_6 respectively.

A CNN architecture is utilized by M_1 , and its main concern is enhancing the accuracy and efficacy of the models. There are a total of 237 layers inside M_1 , which is comprised of seven blocks. In this particular architecture, several different modules are employed, and these modules work together to produce sub-blocks. The dimensions of the model are determined by three separate parameters. These three measurements are known as the width, depth, and resolution. The depth parameter determines the depth of the network, whereas the width parameter represents the total amount of neurons in each layer. The resolution parameter specifies the size of the dataset that will be used to train the model.

M_2 is an effective DL model that makes use of dense interconnection among layers to enhance the performance of

CNNs [57]. Every dense block uses the result of the preceding block's calculation as an input, in addition to using the results of all of the preceding blocks' calculations. This advanced structure has shown significant gains in performance throughout an extensive range of applications of computer vision, involving image categorization, recognizing objects, and the process of segmentation among others. M_2 is a CNN architecture that consists of 201 layers deep. The fundamental concept that drives the design of the M_2 structure is that the feature maps that are generated by each layer can be chained together to create the input for a subsequent layer.

The VGG presented its model that indicated that CNN's reliability is greatly affected by the network's depth [58]. M_3 structure is made up of six distinct phases, the first two of which consist of two conv layers and one layer of max-pooling. In the subsequent three phases, 3 layers of convolutional and 1 layer of max-pooling are created, then 3 FCLs in the final stage. M_4 is significantly more advanced instead of M_3 , it contains 16 conv layers, five layers of max-pooling for obtaining features, and three FCLs for categorization. The M_5 CNN structure is a member of the Inception family [59]. The primary objective of the development of M_5 was to decrease the amount of computational resources that were consumed by changing the earlier versions of the Inception architecture. Within the same level, M_5 makes use of a variety of filter sizes. The network expanded in terms of its width as opposed to its depth, which has the advantageous effect of preventing it from being simply overfitting. M_5 is a combination of 11 distinct inception modules, each of which utilizes convolution layers, pooling layers, as well as the maximum and average value. M_6 is a CNN architecture that was created specifically for mobile or cost-effective devices, and it decreases the expense and size of networks. There are a total of twenty-eight layers in the M_6 architecture, with just one FCL and twenty-seven conv layers being utilized. About the size of the convolutional layer, there are 13 Depthwise Conv that are 3×3 , 1 Conv that is 3×3 , and 13 Conv that are 1×1 .

F. MODEL EVALUATIONS

In the present research, we tackle the multi-categorization issue, such as tooth decay, non-periodontal, and periodontal cases, thus a confusion matrix was constructed for assessing the model's effectiveness. The effectiveness of our proposed model was measured by employing six metrics such as accuracy (ACU), precision (PRE), sensitivity (SEN) (which is referred to as recall (REC)), F1-score, loss, and the Roc curve. The following formulas were used to calculate these metric values, which are provided in Equations 1–8).

$$ACU = \frac{TP + TF}{TP + FN + FP + TN} \quad (1)$$

$$REC = \frac{TP}{TP + FN} \quad (2)$$

$$PRE = \frac{TP}{TP + FP} \quad (3)$$

TABLE 5. Contrasting the Proposed Model with baseline classification methods.

Classifier	ACU	PRE	REC	F1 score	AUC
M ₁	85.94%	87.19%	84.73%	82.67%	97.41%
M ₂	94.37%	94.37%	94.37%	94.45%	99.63%
M ₃	91.96%	91.96%	91.96%	92.28%	99.18%
M ₄	93.57%	93.57%	93.57%	93.23%	99.25%
M ₅	89.95%	90.65%	89.55%	90.16%	98.95%
M ₆	94.77%	94.77%	94.77%	94.71%	99.39%
Proposed Model	84.94%	84.94%	84.94%	83.33%	88.70%
(Without SMOTE TOMKEK)					
Proposed Model	98.39%	98.39%	98.39%	98.31%	99.79%
(With SMOTE TOMKEK)					

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

$$FPR = \frac{FP}{FP + TN} \quad (5)$$

$$TPR = \frac{FP}{FP + FN} \quad (6)$$

$$Loss = y - \bar{y} \quad (7)$$

$$L_{CE} = - \sum_{n=1}^k (L_i \log(p_i)) \quad (8)$$

IV. RESULTS AND DISCUSSIONS

This section examines the suggested approach to the most recent neural network models. Table 5 displays analyses of the model that was suggested and six baseline classification methods such as M₁, M₂, M₃, M₄, M₅, and M₆. To accurately determine the success rate of neural networks with deep connections, we implemented the same parameters to all of them.

A. EXPERIMENTAL SYSTEM

The study experiments were executed using a personal computer that had a storage capacity of 32 gigabytes (GB), a GPU with 11 gigabytes (NVIDIA), and Windows 10 OS. The model was checked out by applying the test set that was created by separating the dataset before the first use of the architecture. The efficiency of an architecture from many different angles can be assessed via a combination of several metrics. The use of the Python programming language has made the creation of new methods, and the Keras library has been put to use to execute six different classifiers. These eight classifiers are made up of six different baseline classification techniques (M₁, M₂, M₃, M₄, M₅, and M₆), the proposed model before up-sampling, and the proposed model through up-sampling.

B. ACCURACY OF PROPOSED MODEL TO OTHER CLASSIFICATION TECHNIQUES

Accuracy is used as the primary metric for evaluating models, the proportion of instances of data that have been accurately categorized about the whole amount of data instances.

Making use of the same dataset, we assessed our suggested PDCNET method with six recent classification techniques, including M₁, M₂, M₃, M₄, M₅, and M₆.

The SMOTE TOMKEK technique provides important outcomes for the suggested model. Although implemented on the same dataset, the model that was suggested via up-sampling, the model that was suggested before up-sampling, M₁, M₂, M₃, M₄, M₅, and M₆ resulted in accuracies of 98.39%, 84.94%, 85.94%, 94.37%, 91.96%, 93.57%, 89.95%, and 94.71%, respectively. Implementing the SMOTE TOMKEK to the suggested approach caused a substantial advancement in performance, as demonstrated in Figure 5.

C. LOSS OF PROPOSED MODEL TO OTHER CLASSIFICATION TECHNIQUES

In a DL model, a loss function is an expression in mathematics that estimates the distinction among the actual as well as the predicted values. It evaluates the efficacy of the architecture and leads the procedure for optimization by offering feedback on the extent to which the model matches the data. This study employs the categorical cross-entropy technique to get an accurate estimate of the overall amount of loss. The cross-entropy function makes use of a loss function to deal with a categorization issue containing multiple classes.

Although executed on the same dataset, the model that was suggested via up-sampling and the model that was suggested before up-sampling resulted in loss values of 0.0561, and 0.5413, respectively. We contrasted our suggested approach to six popular classification techniques such as M₁, M₂, M₃, M₄, M₅, and M₆ with the help of the same dataset. Applying the SMOTE TOMKEK to the suggested PDCNET approach caused a substantial reduction in loss, as demonstrated in Figure 6.

D. PDCNET MODEL ASSESSMENT WITH BASELINE MODELS USING PRECISION

The term “precision” means the level of precise positive assessment relative to the overall quantity of positive evaluations [60]. As previously specified in this study, the approach we propose is a CNN-based model comprised of separate

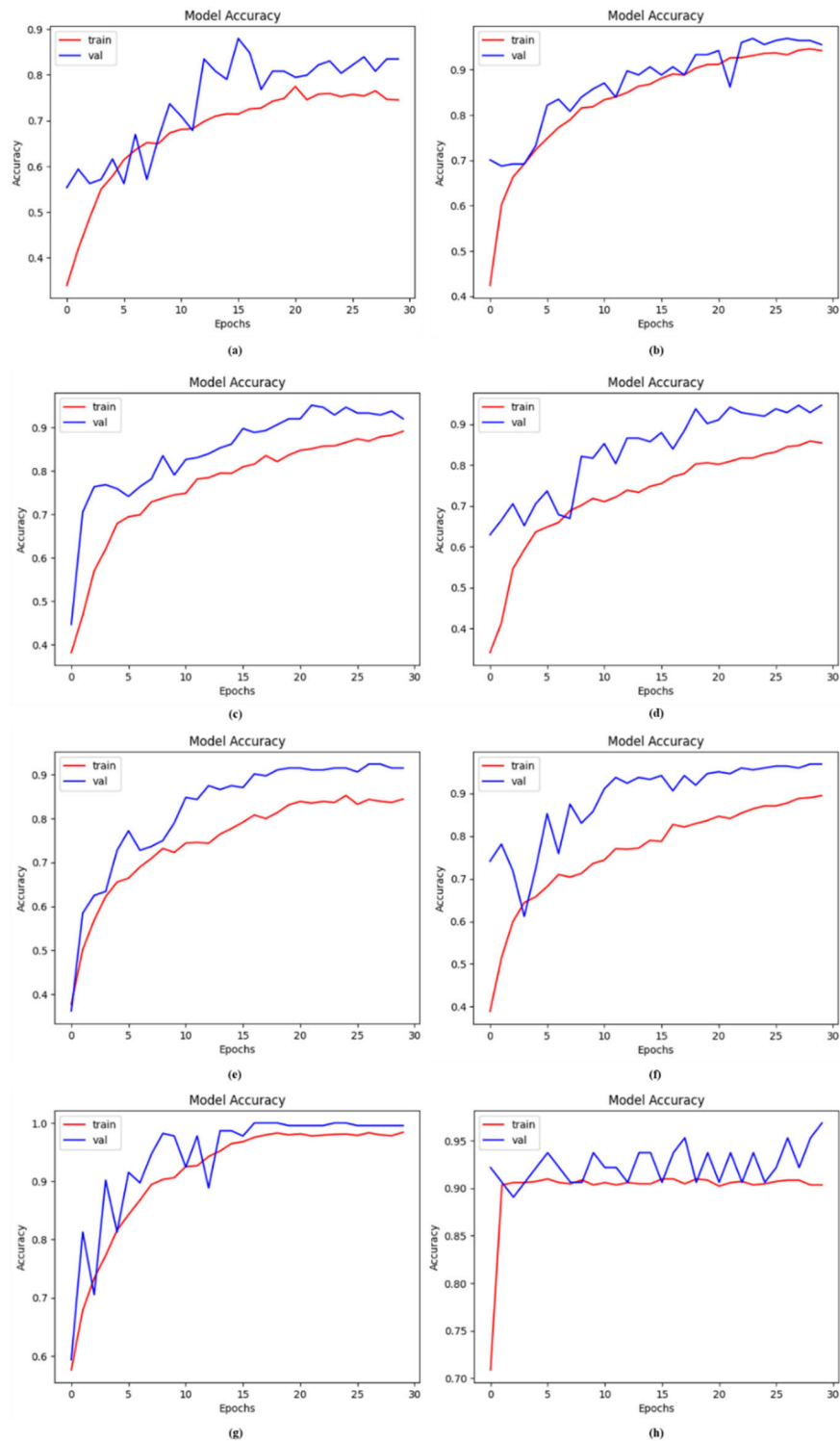


FIGURE 5. Accuracy results of the PDCNET model with six baseline classification techniques; (a) M_1 , (b) M_2 , (c) M_3 , (d) M_4 , (e) M_5 , (f) M_6 , (g,h) PDCNET model with and before up-sampling.

blocks. This model works extremely well for identifying the various dental condition categories because of its effectiveness. On the same dataset, we investigated both our proposed and existing classification techniques, which include M_1 ,

M_2 , M_3 , M_4 , M_5 , and M_6 . In addition, we utilized SMOTE TOMEK to confirm that the dataset was properly and evenly partitioned. As can be seen in Figure 7, the precision of the proposed model that was obtained through up-sampling

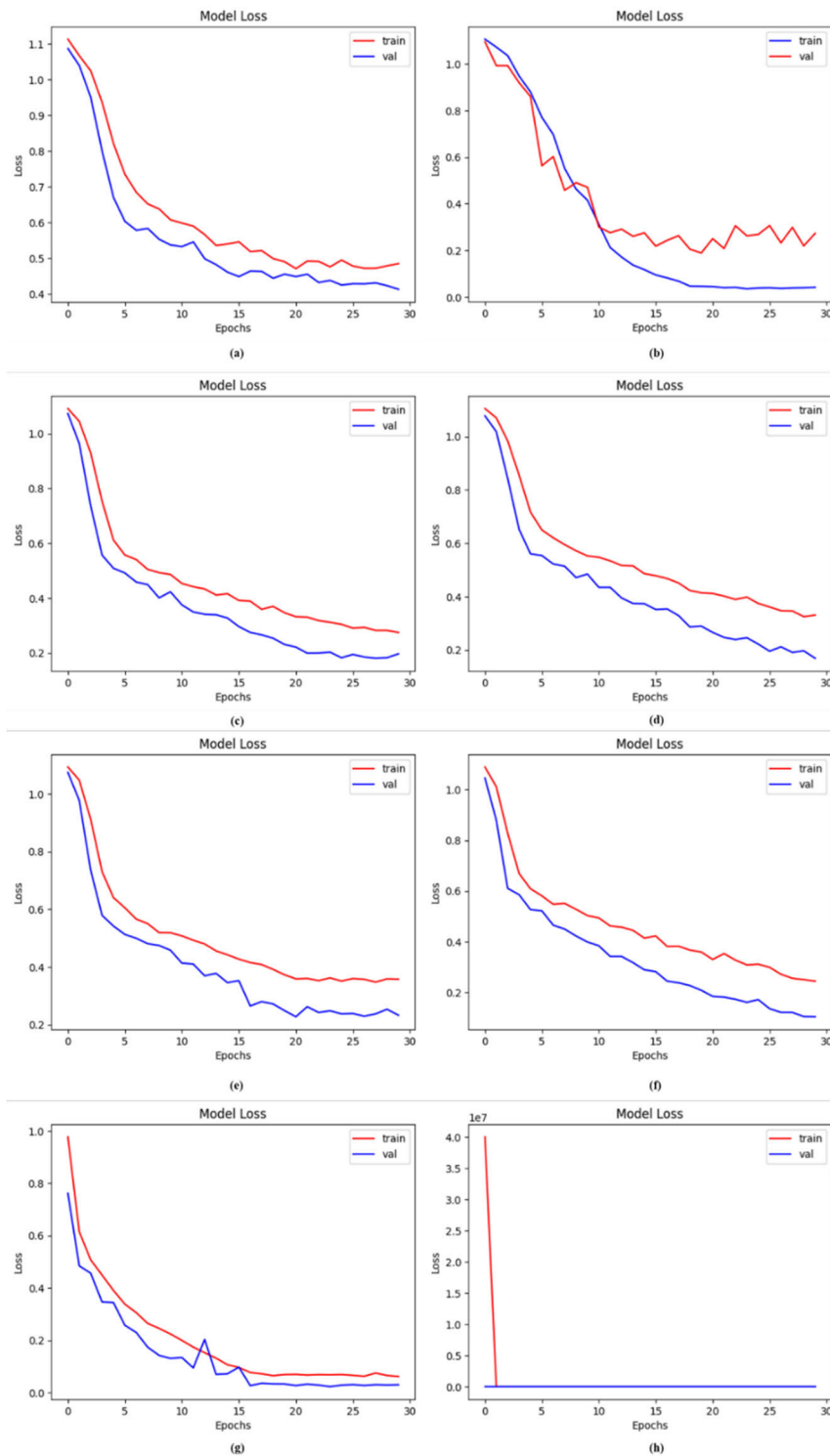


FIGURE 6. Loss values of the PDCNET model with six baseline classification techniques; a) M_1 , (b) M_2 , (c) M_3 , (d) M_4 , (e) M_5 , (f) M_6 , (g,h) PDCNET model with and before up-sampling.

was 98.39%, whereas the precision of the PDCNET model that was obtained before up-sampling was 84.94%. Precision scores for the six different baseline classification methods were as follows: 87.19% for M_1 , 94.37% for M_2 , 91.96% for

M_3 , 93.57% for M_4 , 90.65% for M_5 , and 94.77% for M_6 . The improved effectiveness that can be achieved by combining SMOTE Tomek with the strategy that has been presented is depicted in Figure 7.

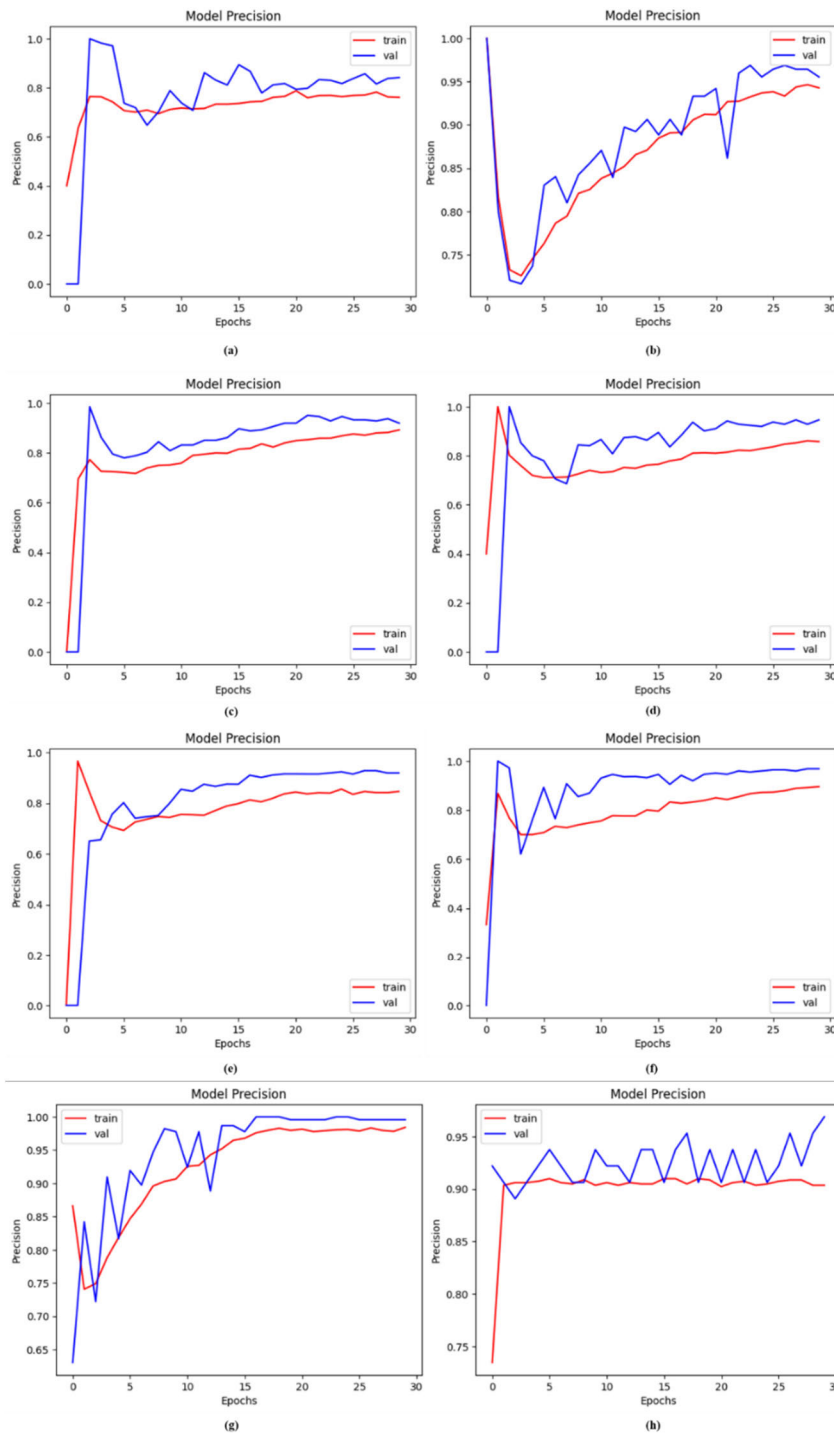


FIGURE 7. Precision results of the PDCNET model with six baseline classification techniques; a) M_1 , (b) M_2 , (c) M_3 , (d) M_4 , (e) M_5 , (f) M_6 , (g,h) PDCNET model with and before up-sampling.

E. ANALYSIS OF PDCNET USING AUC

AUC depicts the possibility that an arbitrary positive instance precedes a random negative instance. The ability of a model to differentiate between multiple categories is evaluated using a metric known as AUC. The architecture that was suggested for this design is based on CNN, and it is comprised of a range of different components, as was described at the beginning of

this analysis. In terms of identifying the various dental disorders classifications, this method works extremely well based on its efficacy. By applying the same dataset, the model that was suggested via up-sampling, the model that was suggested before up-sampling, M_1 , M_2 , M_3 , M_4 , M_5 , and M_6 resulted in AUC values of 99.79%, 88.70%, 97.41%, 99.63%, 99.18%, 99.25%, 98.95%, and 99.39%, respectively. Implementing

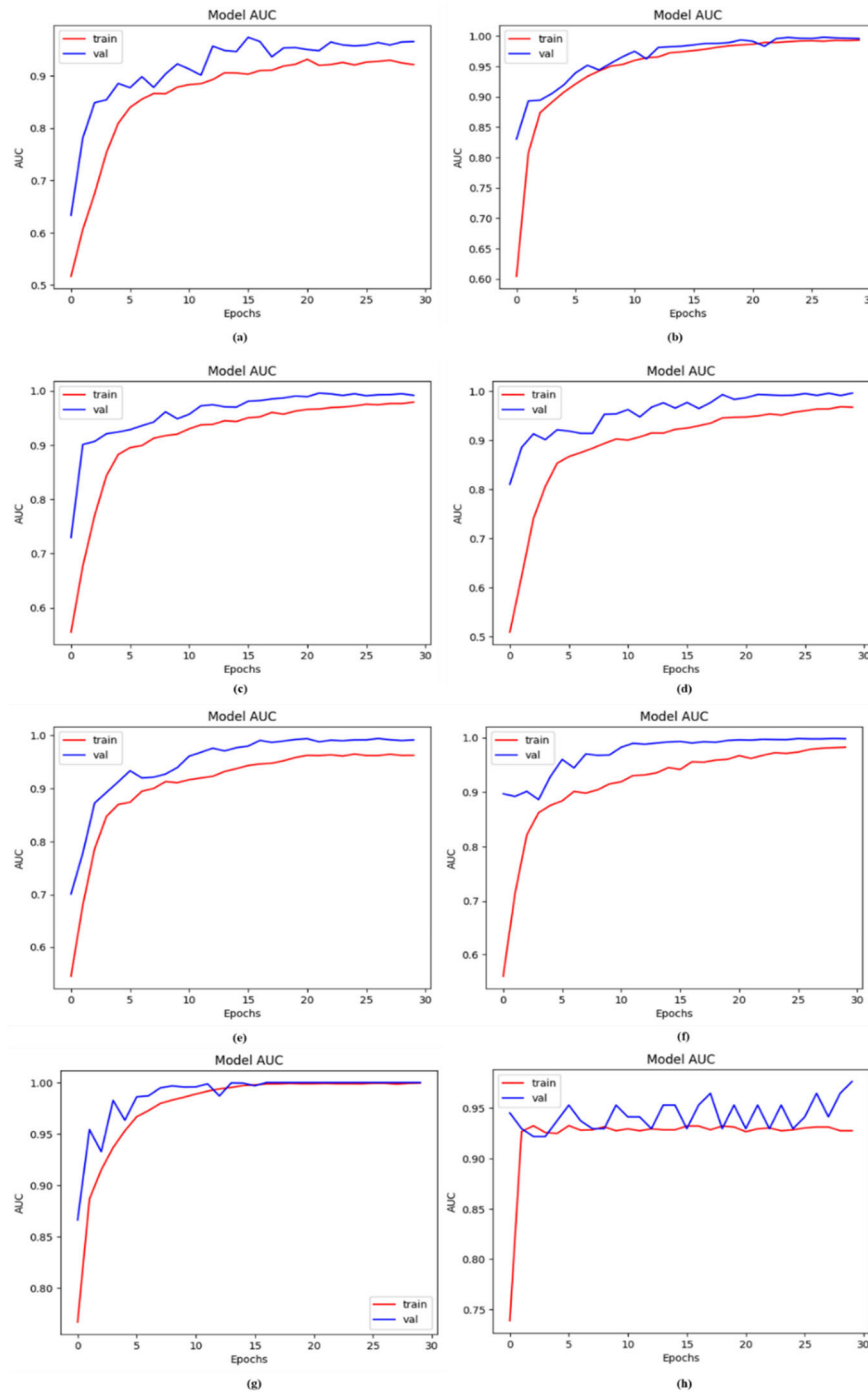


FIGURE 8. AUC results of the PDCNET model with six baseline classification techniques; a) M_1 , (b) M_2 , (c) M_3 , (d) M_4 , (e) M_5 , (f) M_6 , (g,h) PDCNET model with and before up-sampling.

the SMOTE TOMEK to the suggested approach caused a substantial advancement in performance, as demonstrated in Figure 8.

F. RECALL OF PDCNET TO OTHER METHODS

The term “sensitivity score” is used interchangeably with “recall” [61], [62]. It is a comparison of true positive predictions to the real amount of correct positive estimates. A significant recall value suggests a greater proportion of

positive instances have been recognized [63]. Making use of the same dataset, we assessed our suggested method with six recent classification techniques, including M_1 , M_2 , M_3 , M_4 , M_5 , and M_6 . All of these strategies were applied to classify the data. As can be seen in Figure 9, the recall of the proposed model that was obtained through up-sampling was 98.39%, whereas the precision of the suggested model that was obtained before up-sampling was 84.94%. Recall scores for the six different baseline classification methods

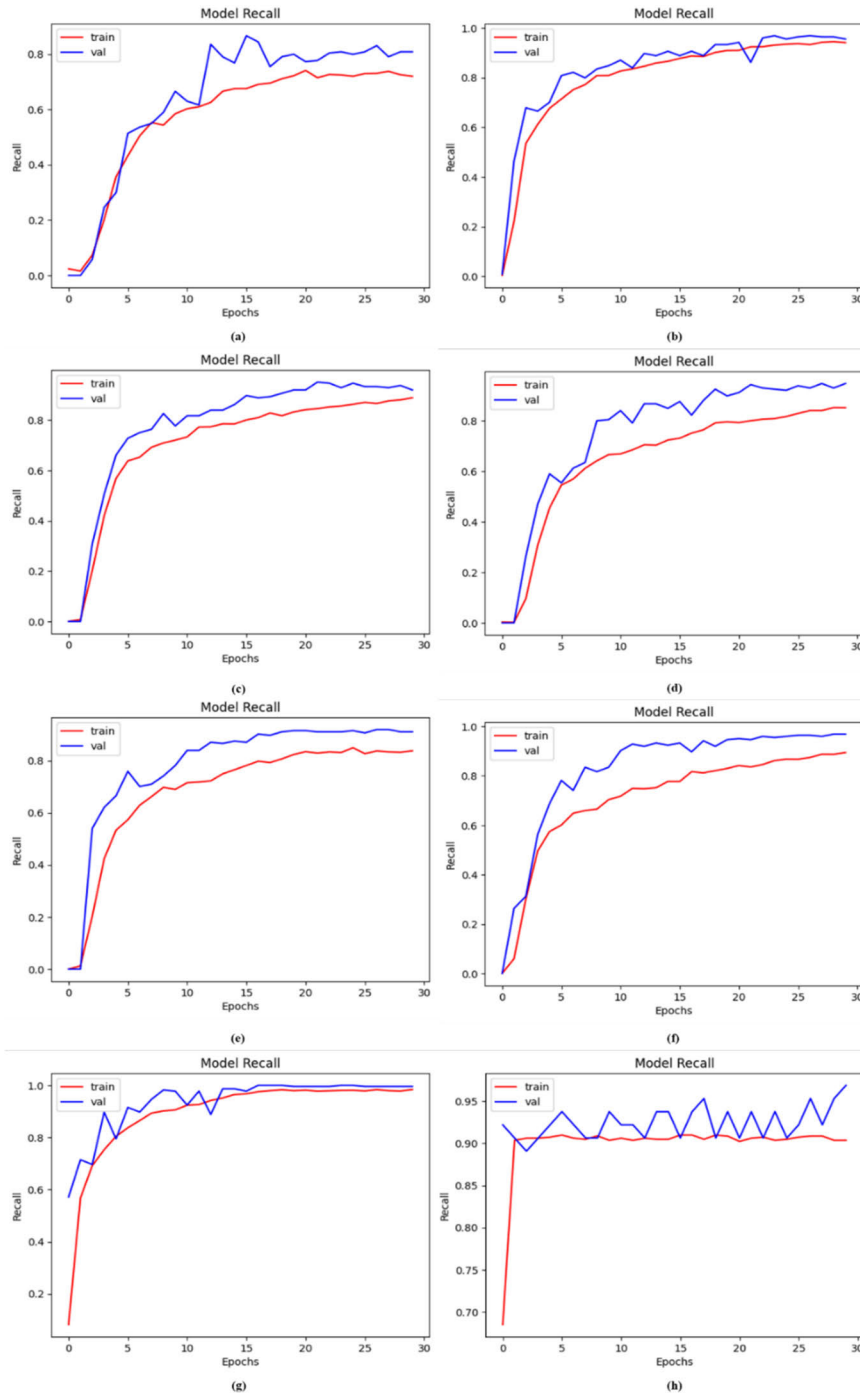


FIGURE 9. Recall values of the PDCNET model with six baseline classification techniques; a) M_1 , b) M_2 , c) M_3 , d) M_4 , e) M_5 , f) M_6 , g,h) PDCNET model with and before up-sampling.

were as follows: 84.73% for M_1 , 94.37% for M_2 , 91.96% for M_3 , 93.57% for M_4 , 89.55% for M_5 , and 94.77% for M_6 . The improved effectiveness that can be achieved by combining SMOTE TOMEK with the strategy that has been presented is depicted in Figure 9.

G. EVALUATION OF THE PDCNET USING F1-SCORE

The F1-Score is a statistic that takes recall as well as precision into evaluation [64]. In general terms, it is defined as the

harmonic mean of both of them. The harmonic mean is simply an alternative method for computing the “average” of a set of values and is typically considered to be more appropriate for proportions (which include precision and recall) compared to the conventional arithmetic mean. Utilizing the same dataset, we examined our suggested approach using six existing classification strategies, namely M_1 , M_2 , M_3 , M_4 , M_5 , and M_6 . As can be seen in Figure 10, the F1-score of the proposed model that was obtained through up-sampling

was 98.31%, whereas the precision of the suggested model that was obtained before up-sampling was 83.33%. F1-score scores for the six different baseline classification methods were as follows: 82.67% for M_1 , 94.45% for M_2 , 92.28% for M_3 , 93.23% for M_4 , 90.16% for M_5 , and 94.71% for M_6 . Implementing the SMOTE TOMEK to the suggested approach caused a substantial advancement in performance, as demonstrated in Figure 10.

H. ASSESSMENT OF PROPOSED MODEL USING EXTENSION OF ROC

A ROC curve is a graphic depiction of the anticipated connection between specificity and sensitivity for every single probable boundary for a set of evaluations [5]. This type of curve is often used to compare the performance of different methods. Analysis is performed among the specificity, which appears on the horizontal axis of the ROC curve, and the sensitivity, which is represented on the y-axis of the graph.

Sensitivity finds the proportion of real positives, while 1-specificity identifies the situation of false positive results [65], [66], [67]. When determining how accurate a predictor is, the area under the curve (AUC) on a ROC graph is an extremely helpful indicator. A greater AUC demonstrates a higher degree of efficacy for the model in general. The purpose of ROC curves is to assess the efficacy of testing methods and, preferred, the reliability of an approach for multiple or binary classes work [10], [11], [12], [13], [14]. ROC curves are commonly utilized in binary categorization as a method for determining how effective a classifier is. Binarization of the generated data is required to broaden the ROC area and curve to accommodate multi-label or classifications with multiple classes [7]. It is feasible to develop one ROC curve for each of the labels; however, each of the different elements of the label assessed structure might also be considered as a binary indicator (micro-averaging) [8].

As shown in Figure 11, a modified version of the ROC curve is applied to assess the model that is suggested using M_1 , M_2 , M_3 , M_4 , M_5 , and M_6 . The up-sampling method caused an important rise in the extension of ROC for the model that was proposed.

I. ANALYSIS OF PROPOSED MODEL USING CONFUSION MATRIX

A confusion matrix is a way of assessing and determining the metrics of multiple categorization models. It encompasses the division of data and all predictions produced by a model over the process of creation or evaluation [57], [58], [59], [60], [61], [62]. To establish the feasibility of the model that we suggested was accurate, we contrasted it with six distinct popular classification methods and then evaluated the findings employing a confusion matrix [63], [64], [65], [66], [67]. Figure 12 is an illustration of the major modifications that have come about to the proposed model as an immediate consequence of utilizing SMOTE TOMEK.

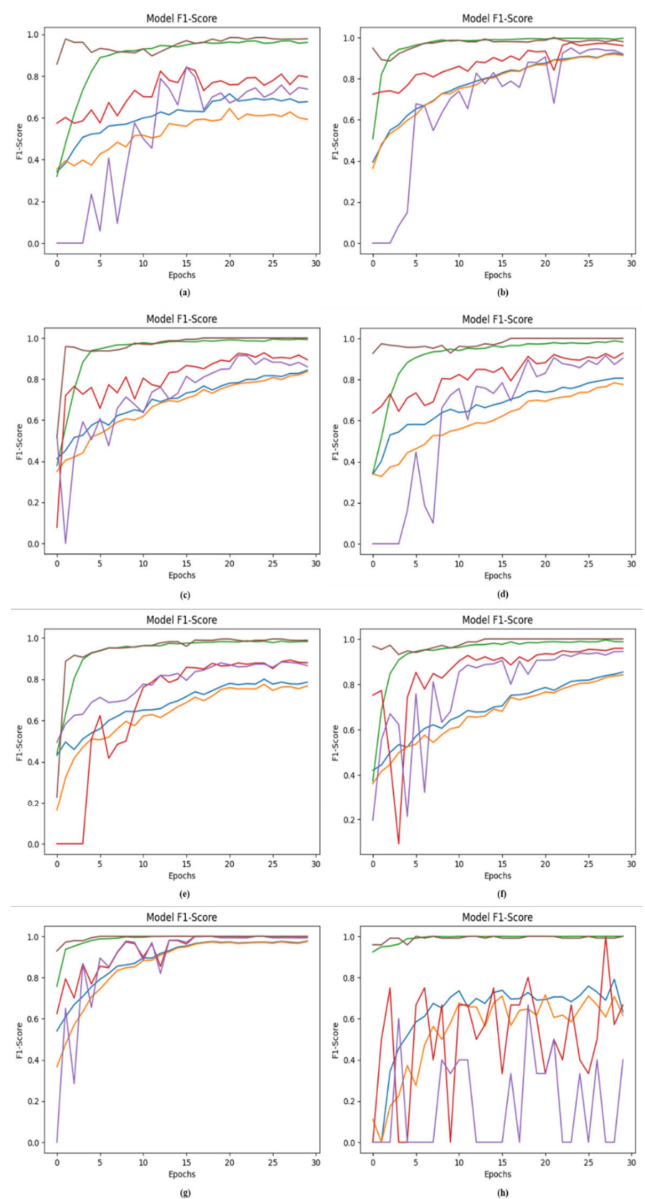


FIGURE 10. F1-score of the PDCNET model with six baseline classification techniques; (a) M_1 , (b) M_2 , (c) M_3 , (d) M_4 , (e) M_5 , (f) M_6 , (g,h) PDCNET model with and before up-sampling.

Using up-sampling, the proposed model correctly identifies 67 out of 69 non-periodontal images, while misidentifying two images as periodontal. In the classification of periodontal cases involving a total of 93 images, 91 were correctly identified as periodontal while two were incorrectly identified as non-periodontal. Figure 12 shows that during the tooth decay classification procedure, 87 out of a total of 87 images were correctly labeled, while none were mis-labeled. In addition, we deployed the Grad-CAM heat-map strategy to give a visual illustration of the findings produced through the model we suggested. The objective of the heat map is to demonstrate and draw attention to the area of the jaw that matters to the classifier. Figure 13 represents the heat map connected with the model that was suggested.

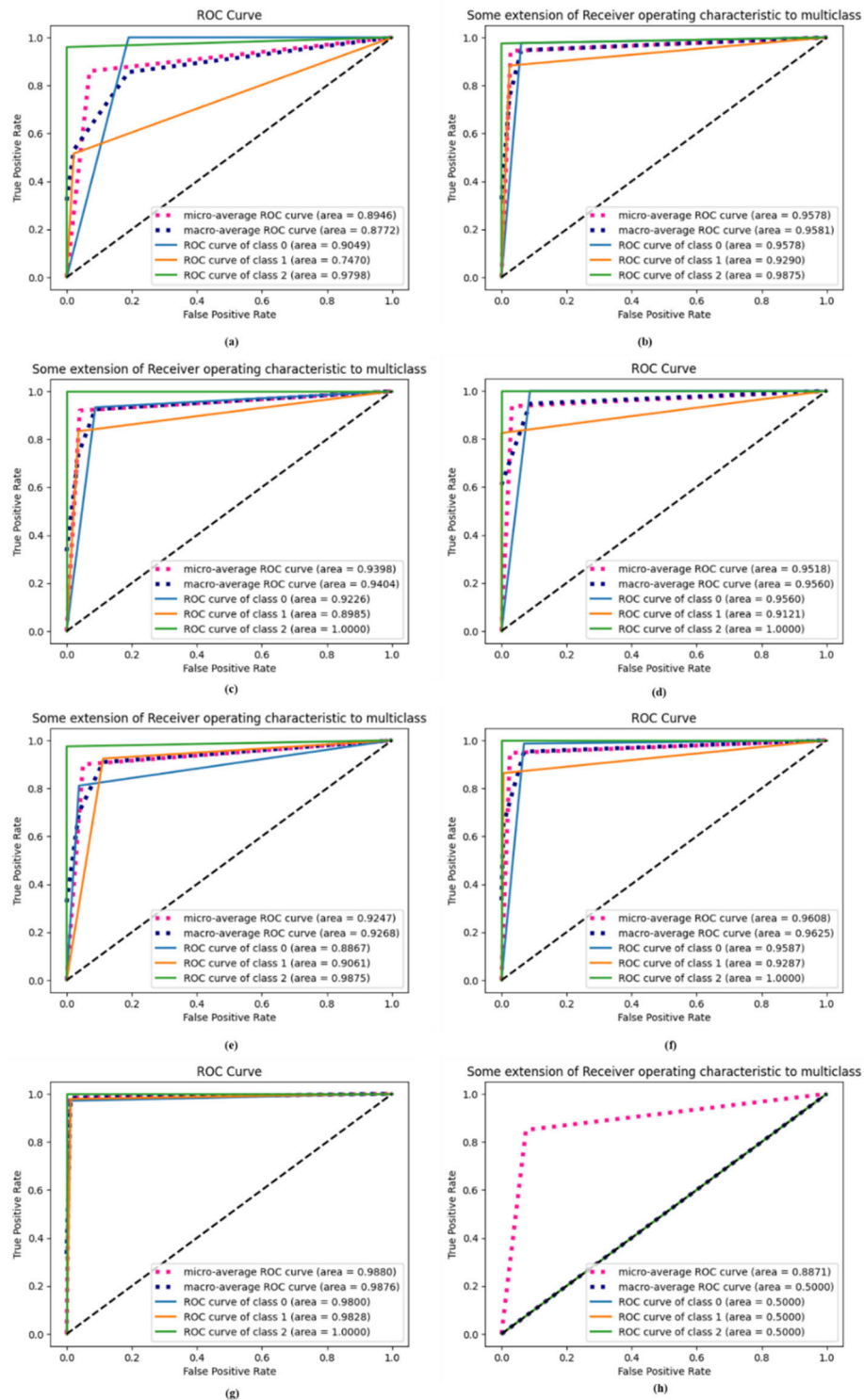


FIGURE 11. ROC extension of the PDCNET model with six baseline classification techniques; a) M_1 , (b) M_2 , (c) M_3 , (d) M_4 , (e) M_5 , (f) M_6 , (g,h) PDCNET model with and before up-sampling.

J. ABLATION STUDY

By incorporating the SMOTE TOMEK approach, this study constructed the suggested PDCNET model. The experimental results were statistically determined using the control

variable approach, which was also used to simultaneously modify a variable to ascertain whether the modified proposed PDCNET model applies to dental problems. This study examined the effectiveness of each PDCNET model in classifying

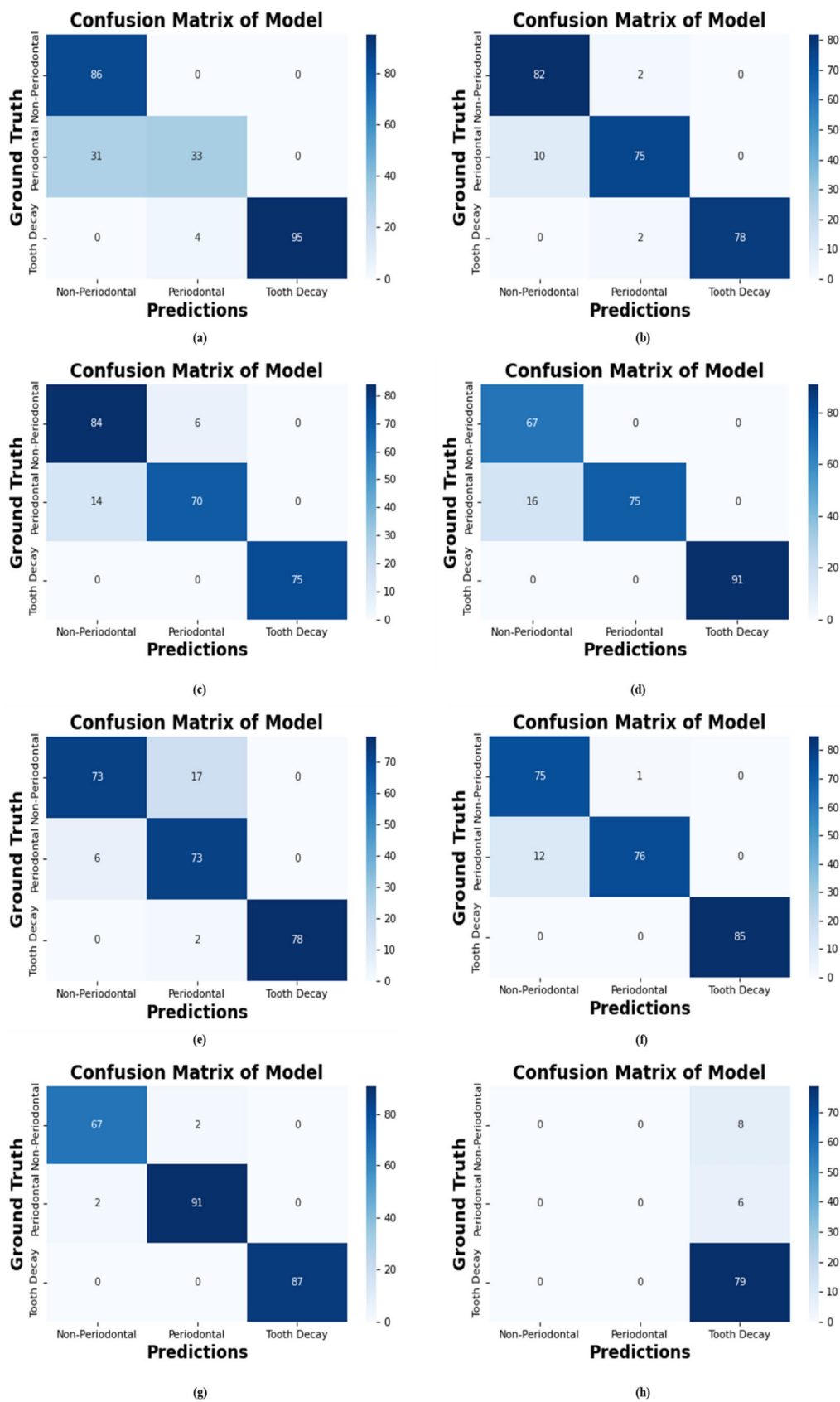


FIGURE 12. Confusion Matrix of the PDCNET model with six baseline classification techniques; a) M₁, (b) M₂, (c) M₃, (d) M₄, (e) M₅, (f) M₆, (g,h) PDCNET model with and before up-sampling.

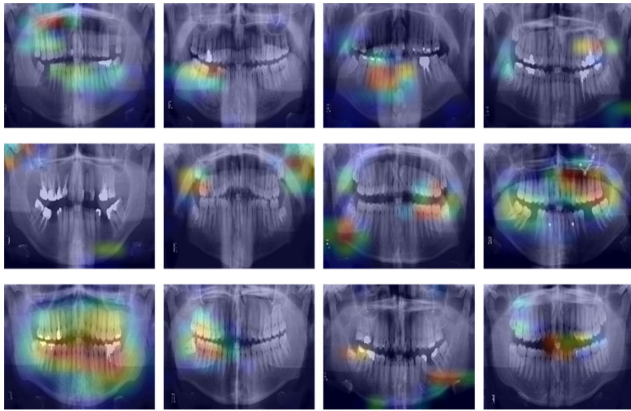


FIGURE 13. Grad-CAM analysis of the suggested model for dental problems.

the three dental diseases both with and without the use of SMOTE TOMEK. We compared the models using the existing methods to determine the importance of the added module to the model. In Experiment 1, the original PDCNET model is demonstrated; in Experiment 2, the SMOTE TOMEK method is introduced with the PDCNET model. Table 6 presents the detailed results of the experiments.

TABLE 6. Results were produced by integrating SMOTE TOMEK with the PDCNET model.

Experiments	PDCNET	SMOTE TOMEK	Image Size	Accuracy
1	✓	×	$150 \times 150 \times 3$	84.94 %
2	✓	✓	$150 \times 150 \times 3$	98.39 %

When the findings of Experiment 1 and Experiment 2 are compared, it is evident that the addition of the SMOTE TOMEK into the PDCNET model leads to an improvement of 13.45% in the model's average classification accuracy of dental disease. The reason for producing enhanced results in Experiment 2 is that SMOTE addresses the class imbalance of the dental disease dataset by producing synthetic dental images for the minority class, while Tomek Links eliminates noisy or overlapping images by removing majority class instances that are close to minority class instances. Therefore, a balanced dental disease dataset helps the proposed PDCNET model to better generalize across both classes, leading to improved performance and reduced bias towards the majority class.

K. COMPARISON OF THE PROPOSED MODEL USING SOTA

In this section, we examine the suggested approach to earlier research [61], [62], [63], [64], [65], [66], [67] that has been conducted in recent years. In addition, the results of these analyses [57], [58], [59], [60] are contrasted directly using the suggested approach that has been developed. Table 7 defines the results of each of the evaluation metrics employed in the suggested approach to the results gathered from recent studies. The evaluation metrics used for the proposed model

include measurements such as accuracy, recall, F1-score, and precision.

L. DISCUSSION

A vast range of disorders that affect the teeth, gums, and other oral structures are included in the category of dental illnesses [4], [5], [6], [7], [8], [9]. When these illnesses go untreated, they can cause several problems, such as pain and discomfort [13]. Gingivitis is a disorder characterized by gum inflammation, usually caused by plaque accumulation. Gum inflammation, redness, and bleeding are a few of the signs [21], [22], [23]. However, gingivitis can be reversed with proper dental hygiene. Gingivitis may progress to periodontitis, a more severe form of gum disease, if treatment is not received [25]. The condition known as periodontitis is characterized by inflammation and infection of the components that support the teeth, including the bone. If it is not handled, it might result in the loss of teeth [37], [38], [39], [40], [41].

Thus, in this study, we designed a novel model named PDCNET based on CNN for the classification of periodontal disease using dental radiographs. For this study, we select the publicly available dataset of tooth decay for the evaluation of the proposed PDCNET model. The dataset contains imbalanced classes, therefore, SMOTE TOMEK was applied to balance the dataset. Additionally, we fixed the size of the dental image to 150×150 for training the proposed model and baseline classifiers. The classification performance of the proposed model is compared with six baseline models such as M_1 , M_2 , M_3 , M_4 , M_5 , and M_6 . The results obtained by using the PDCNET model and the baseline models are presented in Table 2.

Tables 6 demonstrate that the proposed PDCNET model with SMOTE TOMEK is more capable of diagnosing dental disease and extracting the dominant and discriminative patterns in dental radiographs of the balanced dataset, with an accuracy result of 98.39%.

Furthermore, we also compared the classification results of the proposed model with SOTA in Table 7. As can be seen in Table 6, the researchers who worked on the study that was performed by Oztekin et al. [62] examined three of the most common techniques that had been established for the cavities recognition task. DenseNet-121, ResNet-50, and EfficientNet-B0 were the classification techniques that created the experiments. The DenseNet-121 achieves the significant results of 92.00% accuracy. The use of a DL approach of recognition was discussed by Haghanifar et al., [63] to detect dental decay in panoramic photos. The method utilizes transfer learning to discover the pertinent features, and it makes use of a capsule network to create findings that can be used for future prediction. The model that they constructed was able to attain an accuracy of 86.05%. Karaoglu et al. [61] established an architecture called Faster R-CNN that understands, classifies, and describes objects that are present in an image. Their model achieved an accuracy of 92.49%. In the study [64], they constructed a

TABLE 7. Contrasting the PDCNET Model to recent SOTA study results.

Ref	Year	Model	Accuracy	Precision	F1-score	Recall
[1]	2024	CNN	95.30%	93.60%	93.20%	92.10%
[66]	2023	MobileNet-v2	93.00%	86.00%	89.00%	89.00%
[67]	2023	GBDT, RF, LR, SVM, LSTM	95.19%	93.84%	93.79%	88.44%
[61]	2023	Mask R-CNN	92.49%	96.08%	95.87%	95.65%
[62]	2023	EfficientNet-B0, DenseNet-121, ResNet-50	92.00%	×	91.61%	87.33%
[63]	2023	CNN	86.05%	×	×	90.52%
[64]	2022	YOLO-v4 & U-Net	77.00%	77.00%	77.00%	77.00%
[65]	2020	CNN	89.00%	98.00%	93.00%	91.00%
Ours		Proposed Model Using SMOTE TOMEK	98.39%	98.39%	98.31%	98.39%

two-stage DL technique that was effective in locating the dental pulp as well as crucial points by making use of YOLO-v4 and U-Net. They achieved a classification performance of 77.00%. An improved method for identifying cases of tooth decay was proposed by Kang et al. [67] in their study. This method makes use of machine learning classifiers such as GBDT, RF, LR, LSTM, and SVM [68], [69]. The suggested GBDT developed using a reduced feature set obtained the best performance in classification among these classifiers. Its values for accuracy, precision, F1-score, and recall were 95%, 99%, 93%, and 88%, respectively. Additionally, the model that was suggested via up-sampling, the model that was suggested before up-sampling, M₁, M₂, M₃, M₄, M₅, and M₆ achieved accuracies of 98.39%, 84.94%, 85.94%, 94.37%, 91.96%, 93.57%, 89.95%, and 94.71%, respectively, when applied to the same dataset. The result demonstrates that using the proposed PDCNET model with SMOTE TOMEK led to a substantial improvement in performance.

V. CONCLUSION

Tooth decay is a severe public health problem that goes beyond individual medical care and is becoming worse very quickly. Furthermore, reducing future dental decay-related financial consequences requires early detection and prevention of tooth decay. Even so, it can be challenging and error-prone for novice dentists to manually evaluate dental problems. Thus, this study presented a novel PDCNET model based on CNN and evaluated two publicly available benchmark dental datasets. The dental dataset used in this study has the problem of class imbalance. The SMOTE TOMEK method was applied to handle the imbalanced classes of the dataset. The proposed PDCNET has been evaluated with and without using the SMOTE TOMEK method. The GRAD-CAM method was applied to understand the CNN operation by displaying a heat map of the class activation process. The results indicate that the proposed PDCNET with SMOTE TOMEK improves the identification and localization of dental disease with higher accuracy

which is superior compared to other existing techniques. In the future, the federated learning method with the proposed PDCNET model will be used to ensure patient data privacy.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

ETHICAL APPROVAL

There are no ethical implications regarding the public dataset.

INFORMED CONSENT STATEMENT

There are no ethical implications regarding the public dataset.

DATA AVAILABILITY STATEMENT

The dataset used in this study is publicly available.

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